Accident Analysis and Prevention 164 (2022) 106500

Contents lists available at ScienceDirect

Accident Analysis and Prevention

journal homepage: www.elsevier.com/locate/aap

A proactive lane-changing risk prediction framework considering driving intention recognition and different lane-changing patterns

Qiangqiang Shangguan [a,b], Ting Fu [a,b,*], Junhua Wang [a,b], Shou'en Fang [a,b], Liping Fu [c]

[a] The Key Laboratory of Road and Traffic Engineering, Ministry of Education, Tongji University, Shanghai, China
[b] College of Transportation Engineering, Tongji University, 4800 Cao'an Highway, Shanghai 201804, China
[c] Department of Civil and Environmental Engineering, University of Waterloo, 200 University Avenue West, Waterloo, ON, Canada

ARTICLE INFO

Keywords:
Lane-changing risk prediction
Long Short-term Memory (LSTM)
Driving intention recognition
Trajectory data

ABSTRACT

Proactive lane-changing (LC) risk prediction can assist driver's LC decision-making to ensure driving safety. However, most previous studies on LC risk prediction did not consider the driver's intention recognition, which made it difficult to guarantee the timeliness and practicability of LC risk prediction. Moreover, the difference in driving risks and its influencing factors between LC to left lane (LCL) and LC to right lane (LCR) have rarely been investigated. To bridge the above research gaps, this study proposes a proactive LC risk prediction framework which integrates the LC intention recognition module and LC risk prediction module. The Long Short-term Memory (LSTM) neural network with time-series input was employed to recognize the driver's LC intention. The Light Gradient Boosting Machine (LGBM) algorithm was then applied to predict the LC risk. Feature importance analysis was lastly conducted to obtain the key features that affect the LC risk. The highD trajectory dataset was used for framework validation. Results show that the recognition accuracy of the driver's LCL, LCR and lane-keeping (LK) intentions based on the proposed LSTM model are 97%, 96% and 97%, respectively. Meanwhile, the LGBM algorithm outperforms other machine learning algorithms in LC risk prediction. The results from feature importance analysis show that the interaction characteristics of the LC vehicle and its preceding vehicle in the current lane have the greatest impact on the LC risk. The proposed framework could potentially be implemented in advanced driver-assistance system (ADAS) or autonomous driving system for improved driving safety.

1. Introduction

1.1. Background

Risky lane-changing (LC) maneuvers can often trigger turbulences in traffic flow, affecting driving safety and even leading to traffic crashes (Zheng 2014; Chen et al., 2021b; Liu et al., 2021). According to the statistics of the National Highway Traffic Safety Administration (NHTSA), approximately 240,000 to 610,000 LC crashes are reported each year; at least 60,000 people were injured in these LC crashes (Fitch et al., 2009). Besides, as shown in Hou et al. (2015), approximately 5% of traffic crashes and 7% of crash fatalities in the United States are caused by LC each year. In 2019, there were 10,706 traffic crashes caused by dangerous LC and overtaking in China, accounting for 4.32% of all traffic crashes (Traffic Management Bureau of the Public Security Ministry, 2020). Improper or erratic LC maneuvers are also recognized by NHTSA as one of the main causes of traffic crashes (National Center for Statistics and Analysis, 2019). Predicting LC risk is therefore critical for the development of advanced driver-assistance system (ADAS) to improve road traffic safety (Yu et al., 2019).

Since the development of naturalistic driving experiment (Yang et al., 2019; Sarkar et al., 2021; Yu et al., 2021), unmanned aerial vehicle (UAV) and image processing technology (Krajewski et al., 2018), researchers have obtained large-scale and more comprehensive real vehicle trajectory data, which can better understand the microscopic operating characteristics of drivers during lane changes, and then predict the LC risk more accurately (Chen et al., 2019; Chen et al., 2020b).

1.2. Related studies

1.2.1. LC risk analysis

Previous studies on LC risk analysis are mainly divided into three

* Corresponding author at: The Key Laboratory of Road and Traffic Engineering, Ministry of Education, Tongji University, Shanghai, China.
E-mail address: ting.fu.tj@gmail.com (T. Fu).

https://doi.org/10.1016/j.aap.2021.106500
Received 17 August 2021; Received in revised form 14 November 2021; Accepted 15 November 2021
Available online 22 November 2021
0001-4575/© 2021 Elsevier Ltd. All rights reserved.

perspectives, including LC risk quantification, LC risk influencing factors analysis, and LC risk prediction.

As for LC risk quantification, most past studies focused on the risk evaluation of rear-end crashes with a few exceptions that proposed applying LC risk indicators to quantify LC risk. For example, Park et al. (2018) proposed a new LC risk indicator, which comprehensively consider the risk exposure level and risk severity level, to evaluate the crash risk in a LC event. Chen et al. (2019) derived a crash potential index to quantify LC risk embedded in fault tree analysis and cluster algorithm. Besides, the field theory was also applied to assess driving risk during recent years. Huang et al. (2020) proposed a risk assessment model using field theory to quantify the dynamic driving risk during driver's LC process. Chen et al. (2020a) also conducted a LC risk analysis using field theory, and investigated the different risk profiles using time-series clustering algorithm. In addition, taking motion characteristics of the surrounding vehicles into consideration, Joo et al. (2021) proposed a real-time risk assessment model for LC to avoid potential crashes. For more detailed review about surrogate safety measure for LC risk quantification, refer to Arun et al. (2021) and Zheng et al. (2021).

As for LC risk influencing factors analysis, Mahajan et al. (2020a) proposed a whole methodology to evaluate the crash risk of lane-keeping (LK) and LC maneuvers, and then investigated the relationship between LC risk and traffic flow status. The results proved that the speed drop is significantly associated with LC risk. In addition, Chen et al. (2021a) analyzed the influencing factors of LC risk considering the unobserved heterogeneity of LC maneuvers, and the results indicated that the factors related to headway distance and speed difference significantly affect LC risk.

However, so far, only a few studies have been conducted to predict LC risk. Gu et al. (2020) predicted the crash risk of LC vehicles at interchange merging areas using trajectory data collected by UAV. Chen et al. (2019) proposed an integrated framework which including risk labeling, feature selection, resampling method and risk prediction models to predict LC risk. Furthermore, on this basis, a pre-emptive LC risk prediction technique was also developed by Chen et al. (2020b) to predict the LC risk in advance. In addition, Li et al. (2020) proposed a support vector regression model that uses microscopic traffic parameters such as speed, acceleration and gap distance to predict the impact of LC on crash risk.

Although previous studies have conducted insightful investigations on LC risk prediction, significant gaps exist:

- Existing studies on LC risk prediction have not considered the recognition of driver's LC intention, so the timeliness and practicability of the proposed prediction model are limited.
- Existing studies on the LC risk prediction ignore the difference between the two patterns of LC to left lane (LCL) and LC to right lane (LCR).

1.2.2. LC intention recognition

The basic idea of LC intention recognition is to select some indicators that can represent the driver's LC behavior based on prior knowledge, then use supervised learning algorithms for model training, and finally apply the trained model for LC intention recognition (Li et al., 2018). Various supervised learning approaches were employed to recognize LC intentions during the past several years, such as support vector machine (SVM), Bayesian network (BN), Hidden Markov model (HMM), Gaussian mixture model (GMM), and decision tree (DT) model. For example, Kumar et al. (2013) took steering wheel angle and the lateral position of the vehicle as input variables, and adopted a hybrid model which integrates the multiclass SVM model and BN model to predict LC intention. This model can recognize the driver's LC maneuvers 1.3 s in advance. Li et al. (2016) developed a novel algorithm which combines the HMM and BN approaches to recognize the LC intention. In addition, Li et al. (2018) considered the personality traits of the driver, took the lateral position of the vehicle, the use of turn signals, and the driver's gaze behavior as input variables, and established a hybrid model which integrated BN and GMM for LC intention recognition. Meng and Weng (2012) used the DT approach to predict driver's merging maneuver in the work zone, and the results showed that the accuracy of the proposed model was significantly higher than traditional binary logistic regression model.

In addition, with the advancement of machine learning algorithms and computing capabilities, deep learning approaches have also been widely applied to recognize the LC intentions due to their higher prediction accuracy. For example, Xie et al. (2019) developed a data-driven LC model using Long Short-term Memory (LSTM) neural network and deep belief neural network to identify the driver's LC maneuver, and the results showed that the relative distance between the LC vehicle and the preceding vehicle in the target lane significantly affects the driver's LC decision. Han et al. (2019) utilized a neural network embedded in a LSTM layer to predict driver's LC intentions, and then verified the proposed model with naturalistic driving data. Mahajan et al. (2020b) used the LSTM algorithm to recognize the driver's LC maneuver, and verified the transferability of the model on the unlabeled dataset. To improve the prediction accuracy, Cheng et al. (2021) proposed a Seq2-Seq recurrent neural network to predict LC behavior. The model used the speed, acceleration and steering angle as the input variables, and its prediction accuracy could reach 93.5%.

Due to the reliability and practicability requirements of assisted driving systems, more and more researchers use vehicle trajectory data for LC intent recognition (Xu et al., 2019; Pang et al., 2020; Xia et al., 2021; Zhao et al., 2021). For example, Zhao et al. (2021) developed a LC intention prediction model using HMM and naturalistic driving data from Safety Pilot Model Deployment (SPMD) program. Based on the collected driving trajectories using unmanned aerial vehicle, Xu et al. (2019) proposed a hybrid model of GMM-HMM to detect the LC maneuver at off-ramps of freeway. Also, Xia et al. (2021) carried out a human-like LC intention recognition model and verified it with the Next Generation Simulation (NGSIM) vehicle trajectory data. Besides, Pang et al. (2020) developed a LC decision model that considers memory effects and driver heterogeneity, and then validated the model using NGSIM trajectory data.

To sum up, although many LC intention recognition studies have been conducted, the following research gaps are found:

- Most studies on LC intent recognition use deep learning algorithms to improve prediction accuracy. However, the time window length of input features is mostly determined based on prior knowledge (Li et al., 2018; Xie et al., 2019). So, how to choose a suitable time window length to accurately identify LC intention is still worthy of attention.
- Most of the existing intention recognition models are developed based on the driving data at a specific moment before or during the LC, and cannot well capture the characteristics of the driver's entire lane change intention process.

1.3. Research objectives and contributions

To bridge the above-mentioned research gaps, this study aims to develop a framework that includes a LC intention recognition module and a LC risk prediction module to predict the LC risk before the driver starts LC maneuver. Three key steps will be taken in this study:

- An LSTM-based LC intention recognition model will first be developed using vehicle motion features, inter-vehicle interaction features, and driving utility features as input features.
- Based on the results of LC intention recognition model, a LC risk prediction method will be established, which integrates feature extraction, risk labeling, resampling methods, risk prediction models, and prediction performance evaluation, to predict the driving risk of LCL and LCR.

- Considering the risk differences between the two LC patterns, LCL and LCR, the risk influencing factors of these two LC patterns will be analyzed separately.

The contributions of this study can be drawn as follows:

- A proactive lane-changing risk prediction framework will be proposed, which considering driving intention recognition and different LC patterns.
- Time-series trajectory data of the driving intention phase will be extracted for LC intention recognition, and the optimal time widow length of intention recognition will be obtained through deep learning algorithms.
- The importance of risk influencing factors in LCL and LCR scenarios will be obtained, which can provide a basis for risk control in the LC process.

2. Data preparation

2.1. The highD trajectory dataset

The highD trajectory dataset was employed in this study, which is an open naturalistic vehicle trajectory dataset on Germany highways recorded by drones (Krajewski et al., 2018). As vehicles drive on the right-hand side in Germany, this study mainly focuses on the right-hand traffic, in which the driver usually sits on the left side of the car. The highD dataset recorded the trajectory data more than 110,000 vehicles with a total driving distance of 45,000 km. The recorded trajectory data came from six different locations, and the recording time was approximately 16.5 h. The highD dataset website (http://www.highD-dataset.com) describes a detailed composition of the dataset. In addition, an advanced computer vision algorithm, U-Net, was applied to annotate the static and dynamic vehicles. Besides, a tracking algorithm was used to match the vehicle's track between two frames. At last, an additional track postprocessing approach was carried out to smooth the vehicle's trajectory and the extracted features. As mentioned in Krajewski et al. (2018), the recording frequency of highD dataset is 25 Hz, and the position error of the recording track is less than 10 cm.

2.2. LC and LK events labeling

This study mainly considers three driving maneuvers, namely LK, LCL and LCR. Therefore, these three types of events are labeled for the subsequent research. Fig. 1 illustrates two patterns of LC scenarios, LCL and LCR, where *sub* is the subject vehicle, and *Pre* and *Fol* are the preceding and following vehicles in the current lane, respectively. Besides, *Pre_1* and *Fol_1* are the preceding and following vehicles in the target lane for LCL, respectively. In addition, *Pre_2* and *Fol_2* are the preceding and following vehicles in the target lane for LCR, respectively. The highD trajectory dataset provides the y position of the lane markings y_{LM} and the upper left corner of the detected vehicle y_t, the width of the detected vehicle ω_V, and the laneID where the detected vehicle is located. The t_{LC} in Fig. 1 is the time-stamp when the vehicle's front-center crosses the lane marking, that is, the corresponding timestamp when the laneID changes. The Δt is the time interval between two consecutive frames in the highD database, which is 1/25 s.

Then, according to the y position of the lane marking and detected vehicle, and lane ID of the detected vehicle, these two LC events were extracted as follows:

- **Step 1: Extract the timestamp t_{LC} when the laneID changes and determine the LC type.** Firstly, extract the time stamp when the vehicle changes lane according to the recorded laneID at two adjacent frames. Secondly, determine whether the vehicle changes lane to the left or to the right based on the change of the recorded laneID.
- **Step 2: Extract a whole LC event.** According to the y position of the detected vehicle, the forward–backward search method was used to extract the lane-changing process. As shown in Fig. 1, for any time before t_{LC}, search forward with a time step Δt. Take timestamp $t_{LC} - \Delta t$ in LCR scenario as an example, if the following equation 1

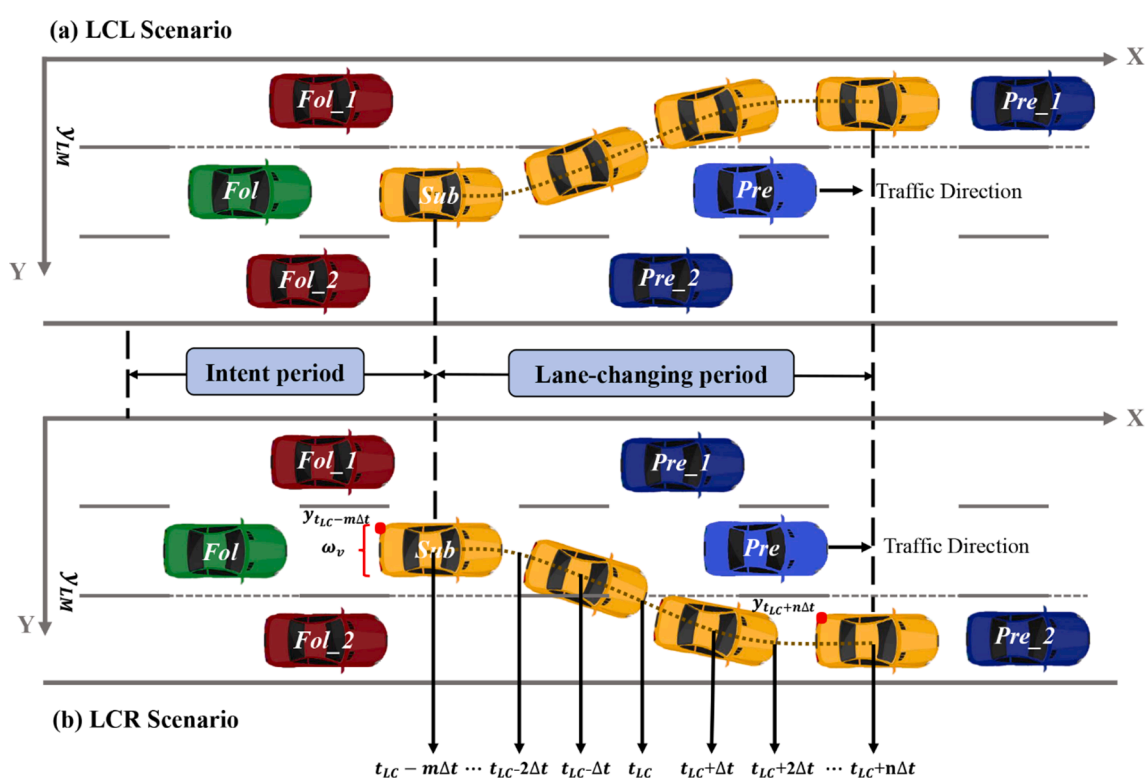


Fig. 1. Illustration of LC scenarios.

holds true, it is determined that time $t_{LC}-\Delta t$ belongs to the entire lane change process. For $t_{LC-\Delta t}$ to $t_{LC}-(m+1)\Delta t$ with a time step Δt, once Equation 2 does not hold, the search will be terminated. Then, the start timestamp of the LC is $t_{LC}-m\Delta t$.

$$\left(y_{t_{LC}-2\Delta t}-y_{t_{LC}-\Delta t}\right)\left(y_{t_{LC}-\Delta t}-y_{t_{LC}}\right)>0 \tag{1}$$

$$\left(y_{t_{LC}-(m+1)\Delta t}-y_{t_{LC}-m\Delta t}\right)\left(y_{t_{LC}-m\Delta t}-y_{t_{LC}-(m-1)\Delta t}\right)>0 \tag{2}$$

where $y_{t_{LC}-(m+1)\Delta t}$, $y_{t_{LC}-m\Delta t}$, $y_{t_{LC}-(m-1)\Delta t}$, $y_{t_{LC}-2\Delta t}$, $y_{t_{LC}-\Delta t}$, and $y_{t_{LC}}$ are the y position of LC vehicle at timestamp $t_{LC}-(m+1)\Delta t$, $t_{LC}-m\Delta t$, $t_{LC}-(m-1)\Delta t$, $t_{LC}-2\Delta t$, $t_{LC}-\Delta t$, and t_{LC}, respectively.

Similarly, as for $t_{LC}+\Delta t$ in backward search process, Equation 3 must be satisfied. For $t_{LC+\Delta t}$ to $t_{LC}+(n+1)\Delta t$ with a time step Δt, once Equation 4 does not hold, the search will be terminated. Then, the end timestamp of the LC is $t_{LC}+n\Delta t$.

$$\left(y_{t_{LC}+2\Delta t}-y_{t_{LC}+\Delta t}\right)\left(y_{t_{LC}+\Delta t}-y_{t_{LC}}\right)>0 \tag{3}$$

$$\left(y_{t_{LC}+(n+1)\Delta t}-y_{t_{LC}+n\Delta t}\right)\left(y_{t_{LC}+n\Delta t}-y_{t_{LC}+(n-1)\Delta t}\right)>0 \tag{4}$$

where $y_{t_{LC}}$, $y_{t_{LC}+\Delta t}$, $y_{t_{LC}+2\Delta t}$, $y_{t_{LC}+(n-1)\Delta t}$, $y_{t_{LC}+n\Delta t}$, and $y_{t_{LC}+(n+1)\Delta t}$ are the y position of LC vehicle at timestamp t_{LC}, $t_{LC}+\Delta t$, $t_{LC}+2\Delta t$, $t_{LC}+(n-1)2\Delta t$, $t_{LC}+n\Delta t$, and $t_{LC}+(n+1)\Delta t$, respectively.

- **Step 3: Remove the incomplete LC events.** Perhaps some drivers will give up changing lanes even when the front center of the vehicle reaches the lane marking. So, more restrictions should be added to eliminate such uncompleted LC events. For example, to ensure the drivers complete their LCR maneuvers, the y position of the *Sub* vehicle at the start and end of the LC should satisfy the following Equation 5 and Equation 6. Same restrictions for LCL events extraction.

$$y_{t_{LC}-m\Delta t}+\omega_v<y_{LM} \tag{5}$$

$$y_{t_{LC}+n\Delta t}>y_{LM} \tag{6}$$

- **Step 4: Extract the LC events include the entire intention period.** Most previous studies about LC intention recognition have mentioned that the intention time is range from 2 s to 5 s (Li et al., 2018; Han et al., 2019; Cheng et al., 2021). So, to guarantee the extracted LC events include the entire intention period, the duration of the intention phase of the extracted LC events should greater than 5 s.
- **Step 5: Extract the trajectory data of vehicles around the LC vehicle.** According to the vehicle's relative position relationship and vehicle ID recorded in the database, the trajectory data of the vehicles around the LC vehicle, including position, speed and acceleration rate, will be extracted. If the vehicle does not exist in a certain position around the LC vehicle, the relative position information is set to infinity, and the speed and acceleration data are filled with the motion characteristics of the LC vehicle.

LK is an event where the driver has no obvious intention to change lanes, such as car-following or free driving. As for LK events extraction, two criteria were applied: *1)* the laneID of the LK vehicle remains constant in the whole detected road segment: guaranteeing the extracted vehicle does not change lanes; *2)* the duration of the LK event needs to be greater than 5 s: ensuring each extracted LK event contains sufficient trajectory data for driving intention recognition. In addition, same with the abovementioned Step 5, the trajectory data of the vehicles around the LK vehicle were also extracted.

Using the highD real vehicle trajectory dataset and above-mentioned extraction algorithms, a total of 1,203 LCL events, 970 LCR events and 2,214 LK events were extracted for further analysis.

3. Methodology

The whole methodology framework of this paper is shown in Fig. 2, which mainly includes the following two parts: i) a driving intention recognition module based on LSTM neural network; and ii) a LC risk prediction module, including dataset construction, risk prediction model, and feature importance analysis.

3.1. LC Intention recognition module

3.1.1. Feature extraction for LC intention recognition

The driver's LC intention is highly complicated due to various complicated factors, such as the position and motion characteristics of the *sub*, the gap distance and speed difference between *sub* and its surrounding vehicles (Xie et al., 2019). Therefore, extracting effective LC intention indicators during driver's intent period (as shown in Fig. 1 is a prerequisite for accurately predicting LC intention. In this study, in addition to the basic features (vehicle type, vehicles' motion characteristics and position information) and interactive features (gap distance, speed difference and acceleration difference between *sub* and its surrounding vehicles), the safety, efficiency, and comfort utility features of the subject vehicle were also extracted as part of the drivers' LC intention indicators. Table 1 presents the summary description of the extracted features.

The *exist or not* in the basic features refers to whether the vehicle exists in a certain position around the LC vehicle. Besides, the safety utility features refer to the longitudinal crash risk between LC vehicle and its surrounding vehicles. The rear-end crash risk indicator (RCRI), an overall measure of the crash probability and crash severity, was employed in this study to represent these safety utility features. Detailed information about RCRI was described in Shangguan et al. (2021a). In addition, to consider the efficiency benefits of the driver executing LC or LK maneuvers, the time headway between *sub* and *pre_1*, the time headway between *sub* and *pre*, and the time headway between *sub* and *pre_2* were extracted as efficiency utility features. Furthermore, drivers tend to maintain a certain speed, and frequent acceleration and deceleration will affect their driving comfort. Therefore, the sum of the squares of the *sub*'s acceleration in X and Y directions was calculated as the comfort utility feature in this study.

3.1.2. LC intention recognition based on LSTM neural network

As a deep learning method for processing time-series data, recurrent neural networks have been widely used in driver intention recognition in recent years (Liu et al., 2019; Xie et al., 2019; Cheng et al., 2021). In this study, LSTM, a neural network model that can solve the gradient disappearance of the recurrent neural network, was applied to recognize the driver's LC intention.

The proposed LSTM model consists of five layers, as shown in Fig. 2. The sequence input layer was fed with the extracted time-series features; the LSTM layer allows information to be selectively passed through gates to add or remove information to the cell state; the fully connected layer was used for dimensional transformation, converting the output results into the recognized driving intentions (LCL, LCR and LK); the softmax layer was used to calculate the probabilities of all possible driving intentions; and the classification output layer was finally employed to output the recognition results. The detailed steps of the proposed LSTM model are shown as follows:

- **Step 1: Input the extracted time series features x_t**

$$x_t=\begin{bmatrix} x_{1,t} \\ x_{2,t} \\ \cdots \\ x_{n,t} \end{bmatrix} \tag{3}$$

where $x_{n,t}$ represents the input feature x_n at timestamp t.

Step 2: Obtain the model parameters through LSTM layer

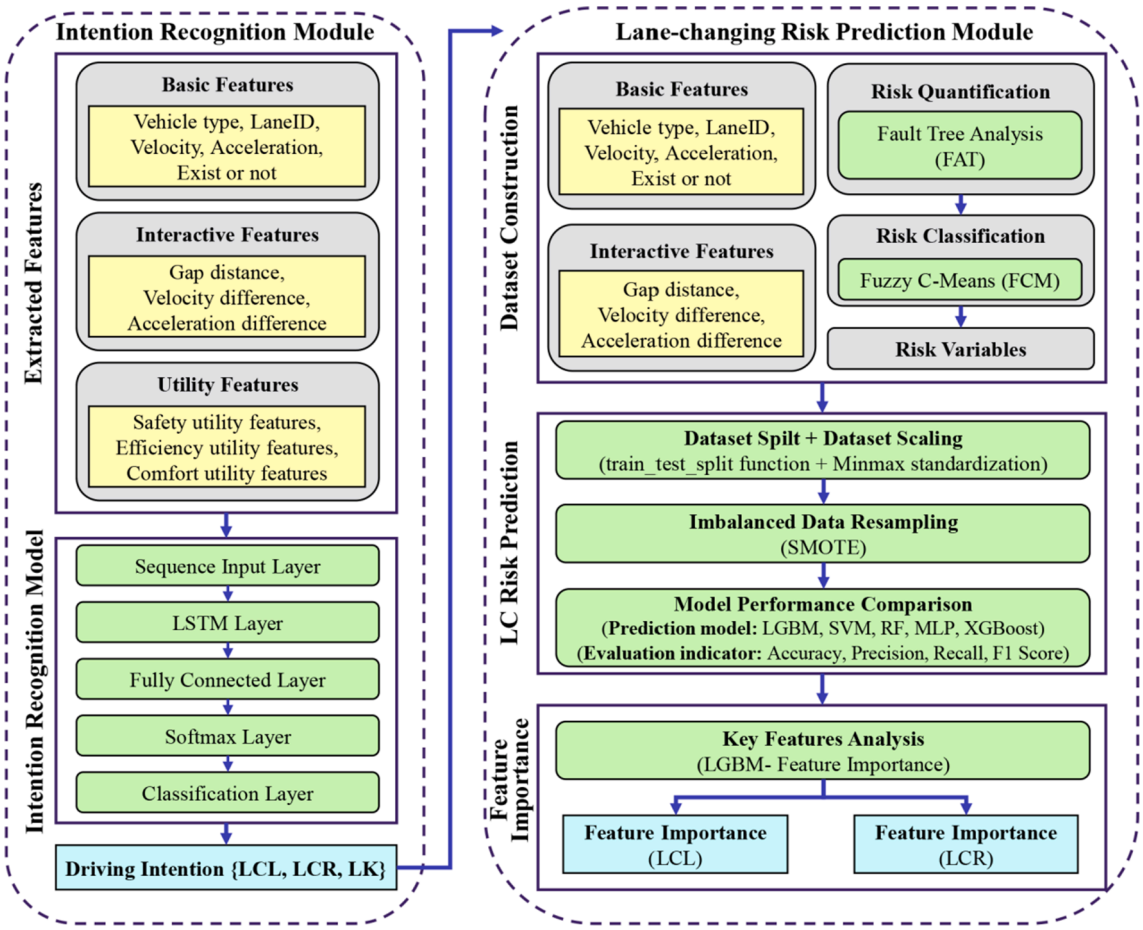


Fig. 2. General methodology framework of this study.

1) Given the input x_t and the hidden state H_{t-1} of the Step 1, the input gate i_t, the forget gate f_t and the output gate o_t can be calculated by the following three equations:

$$i_t = \sigma(W_{xi}x_t + W_{hi}H_{t-1} + b_i) \tag{4}$$

$$f_t = \sigma\left(W_{xf}x_t + W_{hf}H_{t-1} + b_f\right) \tag{5}$$

$$o_t = \sigma(W_{xo}x_t + W_{ho}H_{t-1} + b_o) \tag{6}$$

where W_{xi}, W_{hi}, W_{xf}, W_{hf}, W_{xo}, W_{ho} are weight parameters, respectively; and b_i, b_f, b_o are bias parameters, respectively. All three gates use the sigmoid function σ(.) to control the flow of information in the hidden state.

2) Candidate memory cell C_t, memory cell C_t and the hidden state H_t at the current time step can be calculated by the following equations:

$$C_t = tanh(x_t W_{xc} + H_{t-1}W_{hc} + b_c) \tag{7}$$

$$C_t = f_t \odot C_{t-1} + i_t \odot C_t \tag{8}$$

$$H_t = o_t \odot \tanh(C_t) \tag{9}$$

where W_{xc} and W_{hc} are weight parameters; b_c is bias parameter; and $\odot$ is symbol of Hadamard product.

- **Step 3: Convert the output into the recognized driving intention**

The fully connected layer was then applied to calculate the score of each type of driving intention D_j :

$$D_j = W_j x_t + b_j \tag{10}$$

where W_j and b_j indicate the weight parameter and bias parameter of the driving intention j, respectively.

- **Step 4: Calculate the possibility of each type of driving intention and determine the recognized intention**

The possibility of each type of driving intention, $\widehat{D_j}$, can be calculated using the following equation. Then, the driving intention can be determined according to the probability value of each intention category.

$$\widehat{D_j} = softmax(D_j) = \frac{e^{D_j}}{\sum_{j=1}^{3} e^{D_j}} \tag{11}$$

In this study, 60% of the dataset was randomly selected as the training dataset and input into the proposed LSTM model, 20% of the dataset was applied to select the hyperparameters of the model through validation. Then, using the training dataset and validation dataset, the LSTM model was retrained with the selected hyperparameters. Finally, the remaining 20% dataset was used as the test dataset to verify the LC intent recognition performance. Adaptive moment estimation (ADAM) was employed to iteratively update the weights of the LSTM model, and the initial learning rate was set to 0.01. To avoid the gradient explosion problem of the training model, the gradient threshold was set to 1. Lastly, the other hyperparameters in the LSTM model, including numHiddenUnits, maxEpochs, miniBatchSize, and epsilon were set to 128, 80, 64, and 10e-8, respectively. All input features have been normalized before being fed into the LSTM model. All weight vectors and bias terms are updated during training process. The proposed LSTM model was

Table 1
Summary description of the extracted features for LC intention recognition.

Categories	Extracted features	Conditions	Number of features
Basic features	Vehicle type	Car – 1, Truck – 2	7
	Vehicles' velocity in X direction	Continuous feature (m/s)	7
	Vehicles' velocity in Y direction	Continuous feature (m/s)	7
	Vehicles' acceleration in X direction	Continuous feature (m/s^2)	7
	Vehicles' acceleration in Y direction	Continuous feature (m/s^2)	7
	Exist or not	Not exist – 0, Exist – 1	6
	Lane ID of *Sub*	Road Shoulder – 1,9; Median strip – 5; Vehicle lane – 2,3,4,6,7,8	1
Interactive features	Gap distance between *Sub* and its surrounding vehicles in X direction	Continuous feature (m)	6
	Gap distance between *Sub* and its surrounding vehicles in Y direction	Continuous feature (m)	6
	Velocity difference between *Sub* and its surrounding vehicles in X direction	Continuous feature (m/s)	6
	Velocity difference between *Sub* and its surrounding vehicles in Y direction	Continuous feature (m/s)	6
	Acceleration difference between *Sub* and its surrounding vehicles in X direction	Continuous feature (m/s^2)	6
	Acceleration difference between *Sub* and its surrounding vehicles in Y direction	Continuous feature (m/s^2)	6
Safety utility features	Crash risk between *Sub* and its surrounding vehicles	Continuous feature	6
Efficiency utility features	Headway time between *Sub* and *Pre_1*	Continuous feature (s)	1
	Headway time between *Sub* and *Pre*	Continuous feature (s)	1
	Headway time between *Sub* and *Pre_2*	Continuous feature (s)	1
Comfort utility features	The sum of squares of *Sub*'s acceleration in X and Y direction	Continuous feature (m^2/s^4)	1

conducted using deep network designer toolbox in MATLAB R2019b platform.

3.2. LC Risk prediction module

3.2.1. Dataset construction

(1) Feature extraction for LC risk prediction

Similarly, to guarantee the timeliness of LC risk prediction, this study extracts features from the vehicle trajectory data of the LC intent period. The utility features in Table 1 are mainly used to indicate the benefits that the driver obtains by LC or LK, and are not necessarily related to LC risk prediction. Therefore, only two types of features were extracted for LC risk prediction, including basic features and interactive features. Different from the features extracted in the intention recognition module, the feature extracted for LC risk prediction only include the motion features of the LC vehicle, and the preceding and following vehicles of the LC vehicle in the current and target lane. The detailed descriptions of the extracted features for LC risk prediction are shown in Table 2.

To obtain the optimal time window length of the input features to ensure the accuracy of LC risk prediction, this paper enumerates the time window length from 0.5 s to 5 s, with a step size of 0.5 s. For each time window length, the mean, standard deviation, range, maximum, minimum, 0.5 quantile, kurtosis, and skewness value of the extracted continuous features were calculated as the input features.

(2) Fault tree analysis (FTA) for LC risk quantification

The FTA approach, which commonly used to determine the probability of system failure, was applied in this paper to quantify the LC risk. Fig. 3 shows the fault tree structure of failure to perform a safe LC maneuver. As shown in Fig. 3, the entire system is deemed to have failed if a crash occurs between *sub* and any of its surrounding vehicles during the LC process. The crash risk between *sub* and any of its surrounding vehicles was also calculated using the surrogate safety measure RCRI (Shangguan et al., 2021a).

Then, the LC risk index (LCRI) can be calculated using the following equation:

$$LCRI = 1 - \prod_{i=1}^{4}(1 - RCRI_i) \tag{12}$$

where $RCRI_1$ and $RCRI_3$ are the maximum crash risk between *sub* and its preceding vehicle in the current lane and the target lane, respectively;

Table 2
Descriptions of the extracted features for LC risk prediction.

Categories	Extracted features	Conditions	Number of features
Basic features	Vehicle type	Car – 1, Truck – 2	5
	Vehicles' velocity in X direction*	Continuous feature (m/s)	40
	Vehicles' velocity in Y direction*	Continuous feature (m/s)	40
	Vehicles' acceleration in X direction*	Continuous feature (m/s^2)	40
	Vehicles' acceleration in Y direction*	Continuous feature (m/s^2)	40
	Exist or not	Not exist – 0, Exist – 1	4
	Lane ID of *Sub*	Road Shoulder – 1,9; Median strip – 5; Vehicle lane – 2,3,4,6,7,8	1
Interactive features	Gap distance between *Sub* and its surrounding vehicles in X direction*	Continuous feature (m)	32
	Gap distance between *Sub* and its surrounding vehicles in Y direction*	Continuous feature (m)	32
	Velocity difference between *Sub* and its surrounding vehicles in X direction*	Continuous feature (m/s)	32
	Velocity difference between *Sub* and its surrounding vehicles in Y direction*	Continuous feature (m/s)	32
	Acceleration difference between *Sub* and its surrounding vehicles in X direction*	Continuous feature (m/s^2)	32
	Acceleration difference between *Sub* and its surrounding vehicles in Y direction*	Continuous feature (m/s^2)	32

Note: * indicates the features are the statistical description of the motion feature, including mean, standard deviation, range, maximum, minimum, 0.5 quantile, kurtosis, and skewness.

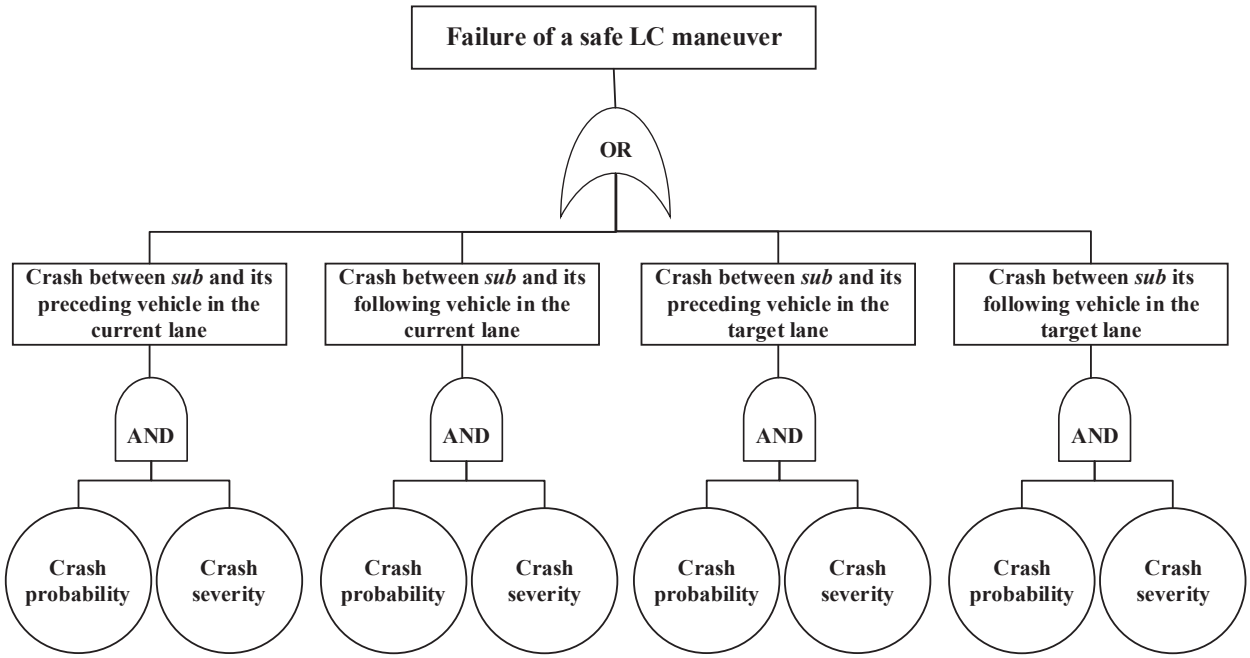


Fig. 3. Fault tree structure of failure to perform a safe LC maneuver.

and $RCRI_2$ and $RCRI_4$ are the maximum crash risk between *sub* and its following vehicle in the current lane and the target lane, respectively.

(3) Fuzzy C-means (FCM) clustering algorithm for LC risk classification

The FCM clustering algorithm, a commonly used unsupervised approach, was employed in this study to classify the LC risk into different levels (Oh et al., 2006; Shi et al., 2019). The principle of the FCM algorithm is to use iterative calculation to maximize the similarity between samples classified into the same category and minimize the similarity between samples in different categories (Bezdek 2013).

As mentioned in Oh et al. (2006), the optimal number of the FCM clusters can be determined by the change of the Wilk's lambda index with different cluster numbers. As shown in Fig. 4, the calculated Wilk's lambda does not decrease significantly when the number of clusters exceeds four, so the optimal classification number of the LC risk is four. Then, input the optimal number of clusters into the FCM clustering algorithm to obtain the classification criteria and classified samples. The FCM clustering algorithm was executed using Fuzzy Logic Designer toolbox in MATLAB R2019b. The detailed process of FCM clustering can refer to Oh et al. (2006). However, due to the limited number of the highest risk level (only 7 samples), this study combines the highest risk samples and the sub-highest category samples. The final classification criteria and sample sizes of the three LC risk categories are shown in Table 3.

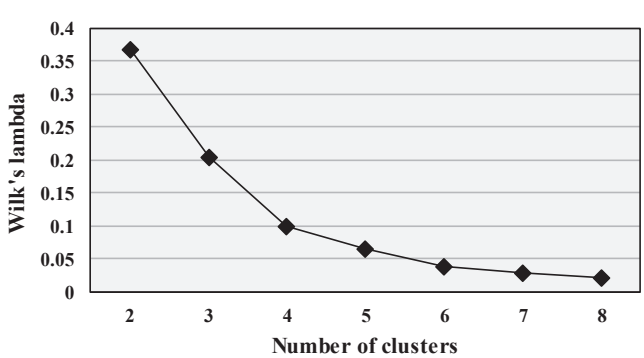


Fig. 4. Wilk's lambda index with different number of clusters.

3.2.2. Light Gradient Boosting Machine (LGBM) algorithm for LC risk prediction

LGBM, a gradient boosting algorithm proposed by Microsoft in 2017, was employed in this study to predict the LC risk due to its accuracy, efficiency, and interpretability (Ke et al., 2017). The LGBM algorithm uses the basic concept of the histogram algorithm to discretize the continuous floating-point eigenvalues into k integers, and construct a histogram with a width of k at the same time. Then use the discretized value as an index to accumulate statistics in the histogram, and finally find the optimal segmentation point. In addition, the LGBM algorithm adopts the leaf-wise growth strategy with depth limitation instead of the level-wise growth strategy, which improves the accuracy of the algorithm and avoids the problem of over-fitting. The detailed process of LC risk prediction based on LGBM algorithm is as follows:

- **Step 1: Split the samples into train set and test set**

The 'train_test_split' function in Sklearn was used to split the samples at a ratio of 0.6:0.2:0.2, that is, 60% of the dataset was employed to training the proposed model, 20% of the dataset was used to validate the trained model to obtain the optimal hyperparameters, and 20% was finally applied to test the trained model.

- **Step 2: Dataset scaling**

Unscaled input features will cause the learning process to be slow or

Table 3
LC risk labelling criteria.

Category	Criteria	LC type	Count (Proportion)
Low-risk	LCRI $\leq$ 0.03	LCL	846 (38.93%)
		LCR	878 (40.41%)
Medium-risk	0.03 $<$ LCRI $\leq$ 0.11	LCL	285 (13.12%)
		LCR	77 (3.54%)
High-risk	LCRI $>$ 0.11	LCL	72 (3.31%)
		LCR	15 (0.69%)

unstable. To eliminate the difference in the range of values between the input features, the minmax standardization method was used to scale the input data.

- **Step 3: Imbalanced data resampling**

The imbalance of the sample size of different LC risk levels in the training set will affect the performance of the prediction model. Therefore, the Synthetic Minority Oversampling Technique (SMOTE) was thus applied to resampling the training dataset (Chawla et al., 2002) to ensure the accuracy of the prediction model.

- **Step 4: Model performance evaluation and hyperparameter tuning**

Model evaluation indicators are the prerequisite for prediction performance evaluation and model hyperparameter tuning. In this paper, four indicators, namely precision, recall, accuracy and F1 score, were selected as model evaluation indicators. Subsequently, GridSearchCV, an exhaustive search method for the specified hyperparameter values of the estimator, was used to select the optimal hyperparameters of the LGBM model using validation dataset. Then, using the training dataset and validation dataset, the LGBM model was retrained with the selected optimal hyperparameters.

- **Step 5: LC risk prediction**

The retrained LGBM algorithm was finally applied to test dataset to predict the LCL and LCR driving risk, respectively. The confusion matrix is also obtained based on the prediction result of test dataset.

Furthermore, to compare the performance of the proposed LGBM classifier and other algorithms, eXtreme gradient boosting (XGBoost), SVM, random forest (RF), and multi-layer perceptron (MLP) were also applied to predict the LC risk.

3.2.3. *Analysis of the LC risk influencing factors*

The importance of the LC risk influencing factors was calculated using the feature importance function of the LGBM classifier. The LGBM classifier provides two approaches for calculating the importance value of the features. One is to calculate the number of times the feature is split in the model, and the other is to calculate the total gains of splits when use the feature. In this study, the total gains of splits when use the feature in the LGBM classifier model were calculated as the importance index. Feature importance is calculated using the 'feature_importances' function of the LGBM classifier in the Sklearn platform, and the type is selected as 'gain'.

4. Results and discussion

4.1. *Results of LC intention recognition*

As mentioned in previous studies, the time window width of the input samples for LC intention recognition is usually in the range of 2 s to 5 s (Li et al., 2018; Han et al., 2019; Cheng et al., 2021). To determine the optimal time window length to recognize the LC intention, the length of the input data $t_I \in [0.5\text{ s}, 5\text{ s}]$ (step at 0.5 s) were selected via comparative analysis conducted by LSTM model. Besides, to explore whether adding interactive features and utility features to input features can improve the accuracy of the proposed model, this study also compared the performance of the prediction model corresponding to the two types of input features.

Fig. 5 shows the training process of the proposed LSTM model. As

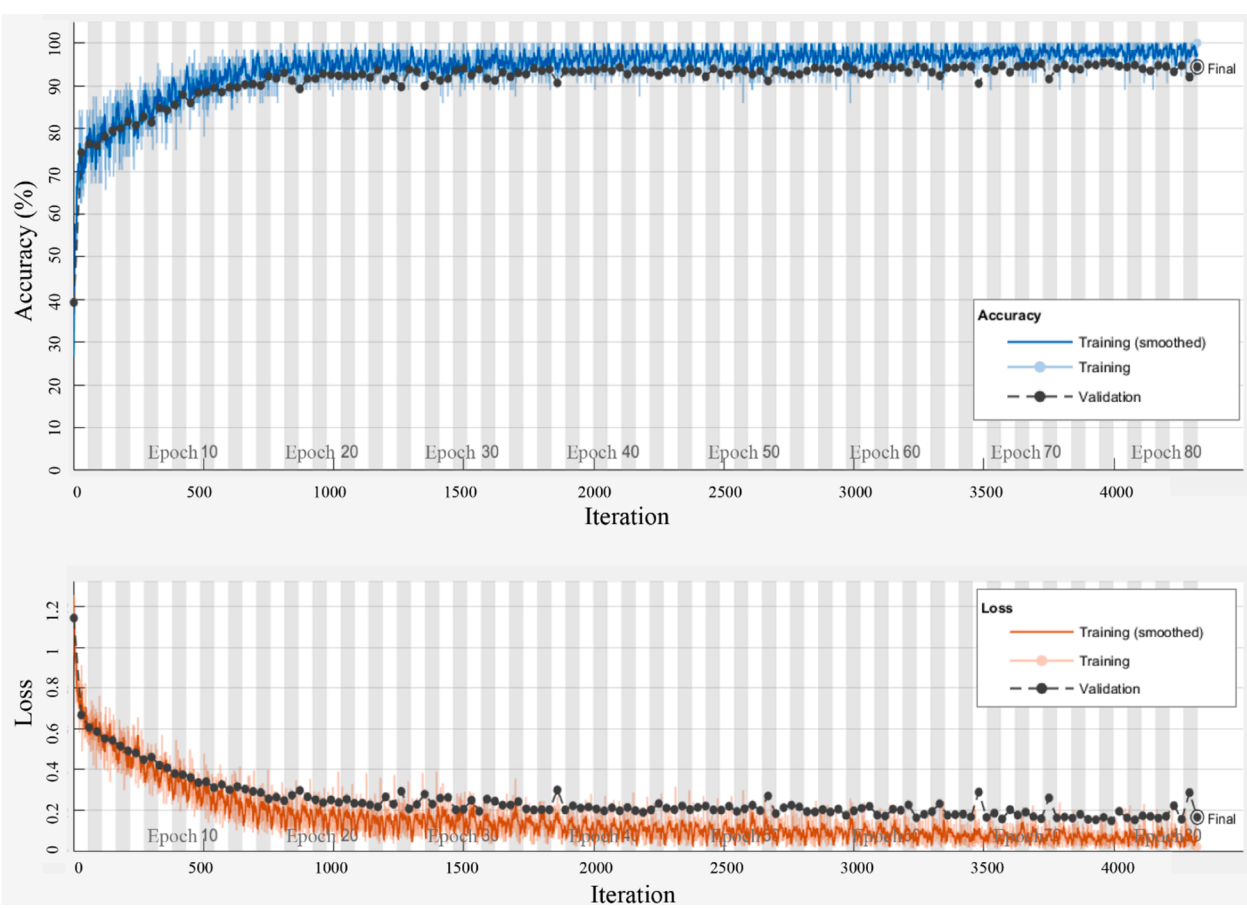


Fig. 5. Training process of LSTM neural network.

shown in Fig. 5, the loss function starts to converge after 30 cycles of training. The prediction accuracy and F1 score of the model corresponding to different input features are obtained after 30 repetitive training and testing, as shown in Fig. 6. It can be obtained from Fig. 6 that the prediction accuracy of the driver's LC intention under different input feature combinations and different time window length can reach more than 90%, indicating that the proposed LSTM model can extract key information from time series trajectory data. Comparing the prediction accuracy and F1 score under different input feature combinations, it can be found that when interactive features and utility features are added to the input features, the prediction accuracy is significantly improved. This also proves the importance of interaction between vehicles and driving utility in driving intention recognition. In addition, when the time window length of the input features is 2.5 s, the accuracy value and the F1 score of the prediction model are both the highest, which is also consistent with the previous studies (Han et al., 2019; Huang et al., 2020; Cheng et al., 2021).

To verify the advancement of the proposed LSTM model, time-series classification model, Convolutional Neural Network (CNN), was employed in this study to recognize the driving intention. Table 4 shows the comparison of the intention recognition results. As shown in Table 4, the accuracy and F1 score of the proposed LSTM model are better than that of the CNN model. When the input features are a combination of basic features, interaction features and utility features and the LSTM model is applied, the prediction accuracy of LCL, LCR and LK are 97%, 96% and 97%, respectively. Fig. 7 presents the confusion matrix of the LC intention recognition results when the input features include basic features, interactive features and utility features, which also shows that the proposed LSTM model has better performance in driving intention recognition.

Table 4
Comparison of intention recognition performance when time window length is 2.5 s.

Model	Input features	LC intention	Accuracy	Precision	Recall	F1 score
CNN	Basic features	LCL	0.94	0.89	0.89	0.89
		LCR	0.93	0.82	0.88	0.85
		LK	0.95	0.96	0.92	0.94
	Overall Accuracy = 0.91 Overall F1 score = 0.89					
	Basic features + Interactive features + Utility features	LCL	0.95	0.90	0.90	0.90
		LCR	0.95	0.87	0.88	0.88
		LK	0.97	0.97	0.97	0.97
	Overall Accuracy = 0.93 Overall F1 score = 0.92					
LSTM	Basic features	LCL	0.96	0.93	0.91	0.92
		LCR	0.95	0.82	0.91	0.86
		LK	0.95	0.97	0.94	0.95
	Overall Accuracy = 0.92 Overall F1 score = 0.91					
	Basic features + Interactive features + Utility features	LCL	0.97	0.95	0.94	0.94
		LCR	0.96	0.92	0.92	0.92
		LK	0.97	0.97	0.98	0.97

Overall Accuracy = 0.96 Overall F1 score = 0.95.

4.2. Results of LC risk prediction and influencing factors analysis

The previous studies on LC are mainly classified as mandatory LC and discretionary LC (Kesting et al., 2007; Ali et al., 2019). However, the difference between the driver's LCL and LCR maneuver has rarely been mentioned. As shown in Fig. 8, the crash risk between LCL and LCR is significantly different ($F = 87.77, p = 0$). Besides, the proportion of high-risk samples in LCL is significantly higher than that in LCR. That is, the crash risk of the driver's LCL maneuver is higher than LCR maneuver. Herein, this study respectively predicts the crash risk of the two LC patterns, LCL and LCR, and analyzes their respective influencing factors.

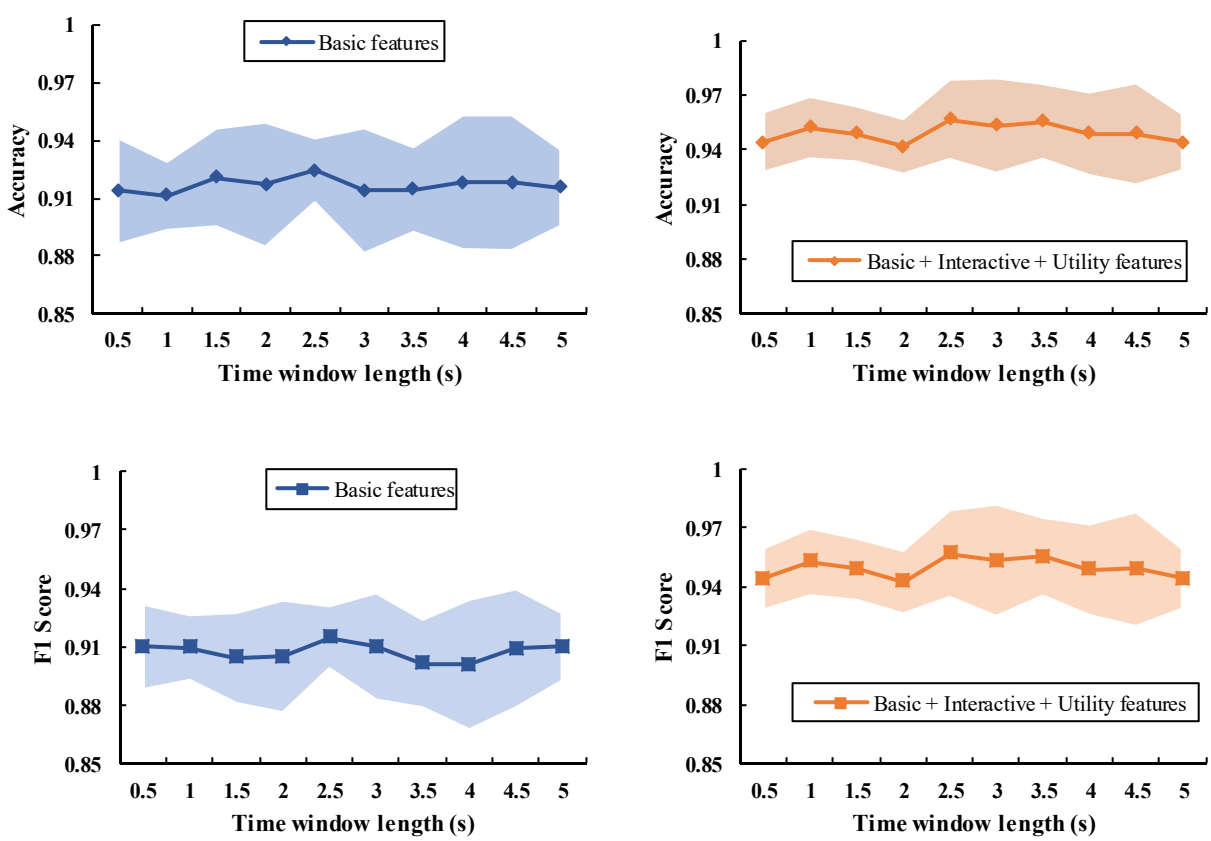


Fig. 6. Accuracy and F1 score of LC intention recognition by time window length under different feature combinations.

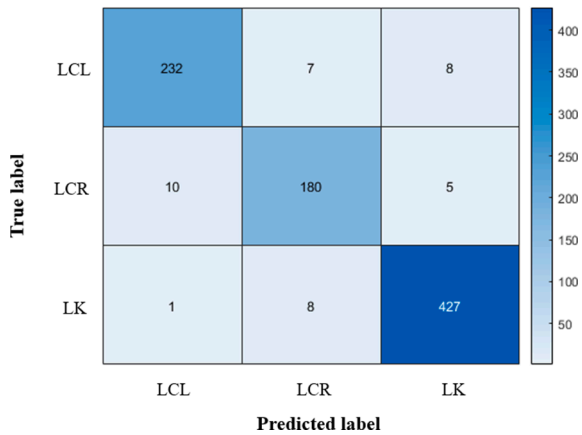


Fig. 7. Confusion matrix of LC intention recognition using LSTM neural network with all features input.

4.2.1. Comparison of LC risk prediction performance

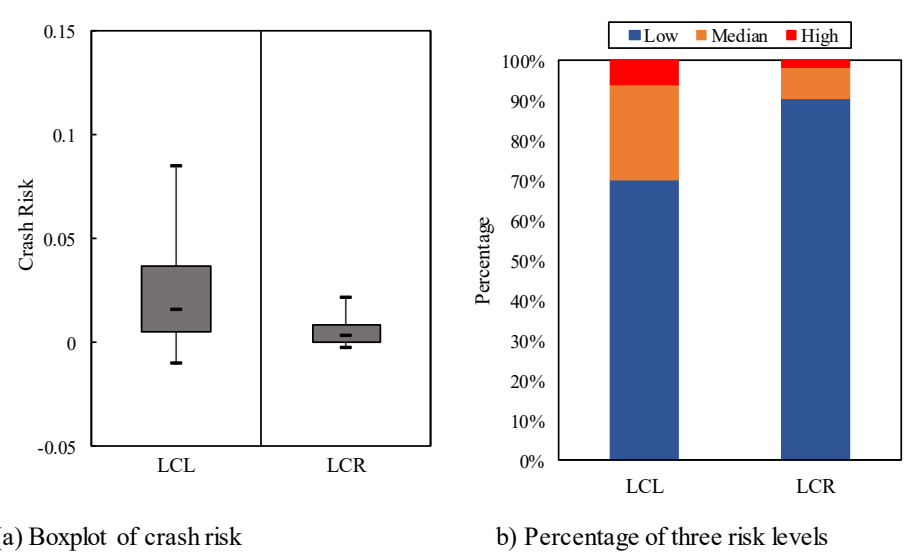
As mentioned above, the time window length from 0.5 s to 5 s with a step size of 0.5 s was enumerated in this study to obtain an optimal time window length of input features to accurately predict the LC risk. Besides, four baseline machine learning approaches, namely XGBoost, SVM, RF, and MLP were also employed in this study to compare the prediction performance with the proposed LGBM algorithm. Fig. 9 shows the accuracy and F1 score of LC risk prediction with different time window lengths and different prediction algorithms. As shown in Fig. 9, the accuracy and F1 score of LCL risk prediction are lower than that of LCR. It may be due to the speed of the vehicles in the left lane is generally higher, leading to more complex driving scenarios for LCL maneuvers, and greater uncertainty in the factors affecting the driver's LCL risk, which makes predictions difficult. In addition, compared to several other machine learning algorithms, the LGBM algorithm achieves higher prediction accuracy. Except that the prediction accuracy of the SVM algorithm increases as the time window length increases, other algorithms are not sensitive to the time window length. As shown in Fig. 9, when the time window length is 2.5 s and the LGBM algorithm is applied, the accuracy and F1 score of LCL and LCR risk prediction reach the highest value.

Table 5 and Fig. 10 present the confusion matrix and the evaluation indicators of the LC risk prediction when time window length is 2.5 s and the LGBM algorithm is used. As shown in Table 5, the risk prediction accuracy of LCL and LCR are 80% and 92%, respectively. Accurately identifying high-risk events can effectively prevent traffic crashes (Shangguan et al., 2021b). In this study, the prediction accuracy of high risk in LCL and LCR is 97% and 99%, respectively. The results show that high-risk samples can be accurately identified, especially high-risk samples in LCR maneuvers. However, the false alarm rate of medium and high risk is still not ideal, possibly because the sample size of medium and high risk is relatively small. Although the oversampling method is adopted, the performance of the LC prediction model is still affected by the imbalance of the dataset.

4.2.2. Analysis of LC risk influencing factors

In this study, the total gains of the splitting when use the feature in the LGBM algorithm were calculated as the feature importance index. The input features can be divided into four categories, including vehicle type, velocity, acceleration and gap distance. Fig. 11 shows the feature importance results of LCL and LCR risk prediction. Four main findings can be obtained as follows:

- Regardless of LCL operation or LCR scenario, the interactive features between *sub* and *Pre* have the highest feature importance. This result shows that during the LC process, the most critical factor affecting the crash risk is the interaction of LC vehicle and its preceding vehicle in the current lane, which is also consistent with Chen et al. (2020b). Besides, among the interactive features between *sub* and *Pre*, the longitudinal gap distance and speed difference between *sub* and *Pre* have a higher importance score. In addition, for LCR maneuver, the lateral speed difference between and the longitudinal relative acceleration between *sub* and *Pre* also have a higher importance score.
- As for LCL maneuver, in addition to the interaction between *sub* and *Pre*, the order of interactive features importance between *sub* and its other surrounding vehicles is *Sub_Fol1* > *Sub_Pre1* > *Sub_Fol*. However, for LCR maneuver, the importance order is *Sub_Fol* > *Sub_Pre2* > *Sub_Fol2*. The results show that in the LCL maneuver, in addition to the motion state of the preceding vehicle in the current lane, the driver should pay more attention to the motion state of the following vehicle in the target lane. For LCR maneuver, in addition to paying attention to the motion state of the front and rear cars in the target lane, the driver also needs to avoid sudden deceleration to ensure the safety of the following car in the current lane.
- As for vehicle types, the type of the LC vehicle has the highest importance score in both two LC patterns, followed by the following



(a) Boxplot of crash risk

b) Percentage of three risk levels

Fig. 8. Crash risk of two LC patterns.

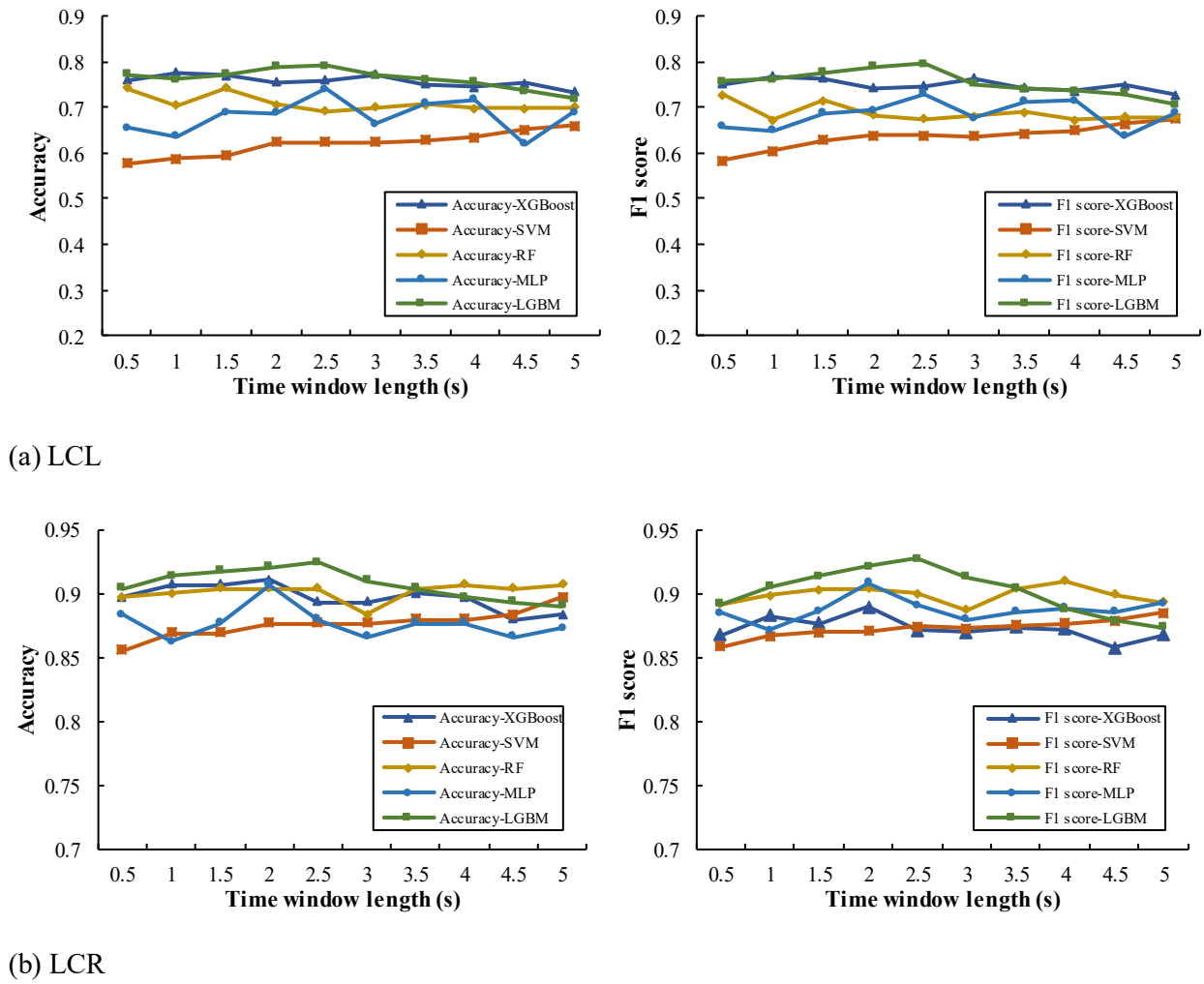


Fig. 9. Accuracy and F1 score of LC risk prediction with different time window lengths.

Table 5
LC risk prediction evaluation indicators using LGBM when time window length is 2.5 s.

LC type	Risk level	Accuracy	Precision	Recall	F1 score
LCL	Low	0.83	0.90	0.85	0.87
	Medium	0.80	0.56	0.64	0.60
	High	0.97	0.75	0.80	0.77
Overall Accuracy = 0.80 Overall F1 score = 0.80					
LCR	Low	0.93	0.98	0.94	0.96
	Medium	0.92	0.50	0.67	0.57
	High	0.99	0.75	1.00	0.86

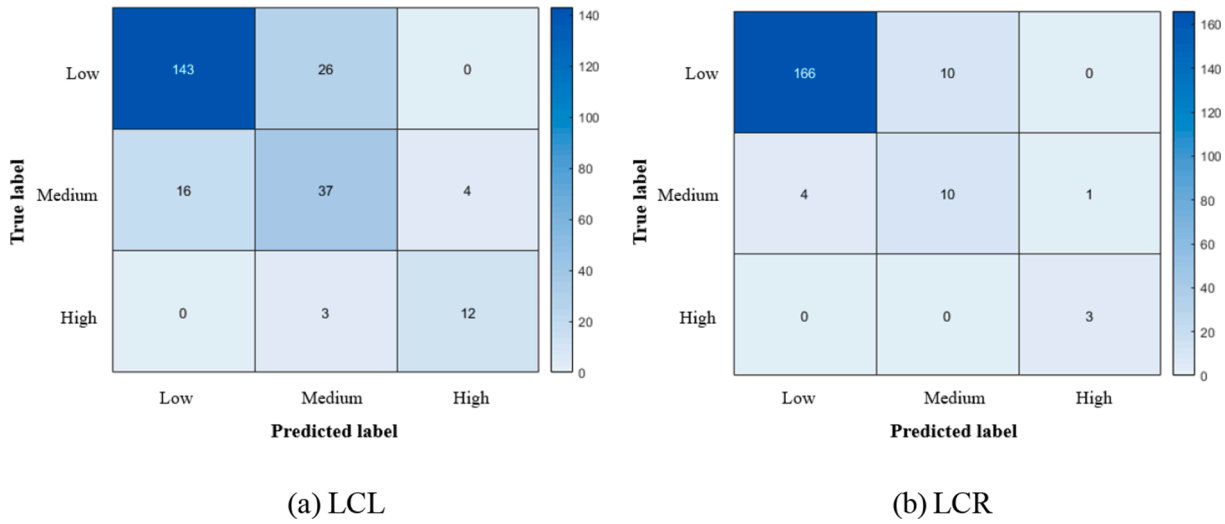
Overall Accuracy = 0.92 Overall F1 score = 0.93.

vehicle in the target lane. It is also interpretable that the crash risks in the LC process are mainly the caused by the maneuver of the LC vehicle and the reaction of the following vehicle in the target lane. Previous studies also mentioned that the fatality rate of the trucks is higher than that of passenger cars (Hyun et al., 2019; Behnood and Al-Bdairi 2020).

5. Conclusion

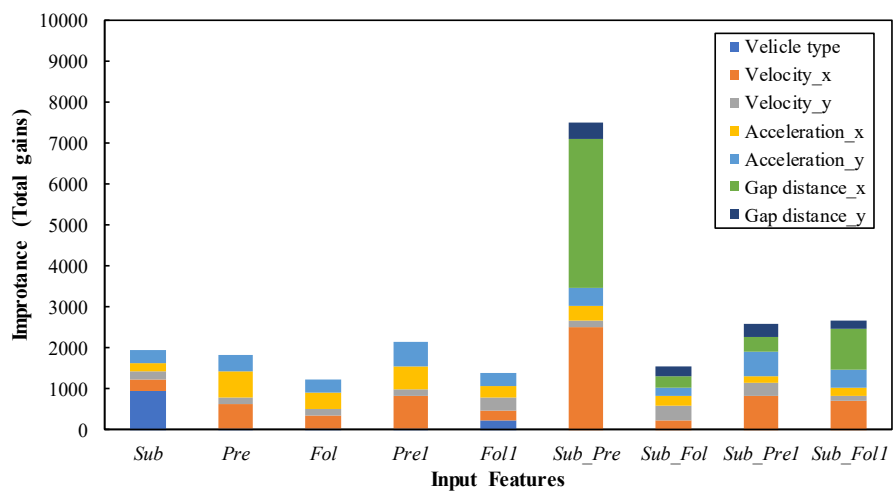
This paper proposed a proactive LC risk prediction framework, which includes a LSTM-based driving intention recognition module and a LGBM-based LC risk prediction module. The proposed framework was evaluated using a large sample of events, including 1,203 LCL events, 970 LCR events, and 2,214 LK events, extracted from an open naturalistic vehicle trajectory dataset from Germany called highD. Three key conclusions can be drawn from our empirical investigation as follows:

- Adding interactive features and utility features to the input features of the driving intention recognition model can improve the recognition accuracy. In addition, the accuracy of the LSTM driving intention recognition model with time-series input is higher than that of the CNN model. When the input time window is 2.5 s, the prediction accuracy of LCL, LCR and LK are 97%, 96% and 97%, respectively.
- The FTA approach, which considering the crash risk between the LC vehicle and its surrounding vehicles during LC process, can more accurately quantify the LC risk. A new gradient boosting algorithm, LGBM, was applied to predict the risk of LCL maneuver and LCR maneuver. In addition, compared with several other machine learning models, the LGBM model is more suitable for LC risk prediction.
- During the LC process, the interaction characteristics of the LC vehicle and its preceding vehicle in the current lane have the greatest impact on the LC risk. In addition, for LCL maneuvers, drivers should pay more attention to the motion state of the following vehicle in the target lane. However, for LCR maneuvers, in addition to paying attention to the state of the vehicles in the target lane, the driver also needs to maintain a safe distance from the following vehicle in the current lane to avoid rear-end collision.

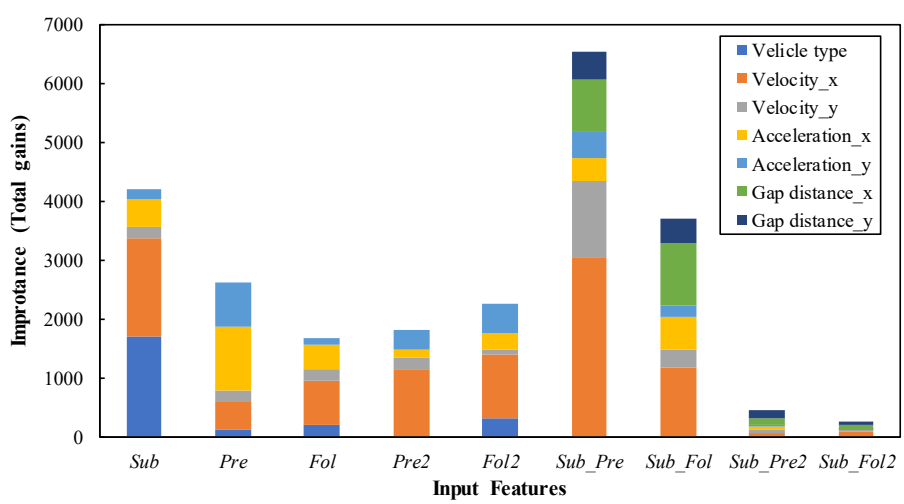


(a) LCL (b) LCR

Fig. 10. Confusion matrix of LC risk prediction using LGBM when time window length is 2.5 s.



(a) LCL



(b) LCR

Fig. 11. Histogram of the feature importance.

As the main contribution, this paper proposed a proactive LC risk prediction framework which integrates the driving intention recognition results and uses the trajectory features of the LC vehicle and its surrounding vehicles in the driver's intention stage to proactively predict the LC risk. The proposed prediction framework can be used to predict the LC risk before the driver starts LC maneuvers. Besides, this study uses real vehicle trajectory data to verify the effectiveness of the proposed framework. The proposed framework could potentially be applied to advanced driver-assistance system (ADAS) to help drivers make driving decisions based on the perception of the surrounding vehicles' motion characteristics and to improve driving safety. In addition, the importance analysis results of LC risk influencing factors can provide theoretical references for driver education and traffic managers. For example, in addition to carefully observing the motion state of preceding vehicle and following vehicle in the target lane, the driver should maintain a safe distance from the preceding vehicle in the current lane during LC. Furthermore, when changing lanes to the right, the driver should try not to take sudden braking maneuvers to avoid a rear-end collision with the following vehicle in the current lane. However, there are still some limitations that need to be improved in the future. For LC risk quantification, only the rear-end crash risk is considered in this study, while the lateral crash risk between vehicles is ignored. As for input features, in addition to the behavioral features extracted from the highD trajectory dataset in this study, many other traffic features, such as weather conditions, light conditions, and road type, have been found in previous studies to have a significant impact on driving intention recognition and risk prediction (Martinez et al., 2017; Li et al., 2020; Das and Ahmed 2021). However, these variables were not included in this study due to the applied trajectory data were obtained from video collected by drone at fixed spots. For example, due to the requirements for the use of drones, the weather conditions and light conditions are mostly sunny and bright, so such features cannot be used as input features of the proposed model. Besides, the used trajectory dataset was recorded on Germany highways, so the road type cannot be applied as the input feature in this study. In future research, the lateral motion characteristics of the vehicle and the driver's steering avoidance behavior will be considered to accurately quantify the LC risk. The recently published TJRD TS trajectory dataset, which includes road alignment information, meteorological information and road conditions at the road-range level, can be used to deeply investigate the driving intention recognition and risk prediction (Wang and Fu 2021). In addition, the correlation between input features needs to be considered in the future work. At last, more artificial intelligence algorithms also need to be adopted to improve the prediction accuracy and reduce false alarm rates.

CRediT authorship contribution statement

Qiangqiang Shangguan: Conceptualization, Methodology, Data curation, Investigation, Software, Writing – original draft. **Ting Fu:** Investigation, Data curation, Writing – review & editing. **Junhua Wang:** Conceptualization, Writing – review & editing, Funding acquisition. **Shou'en Fang:** Conceptualization, Supervision. **Liping Fu:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study was jointly supported by the National Key R&D Program of China (2019YFB1600703), the Chinese National Natural Science Foundation (71871161), and the grant of Science and Technology Commission of Shanghai Municipality (19DZ1202100).

References

Ali, Y., Zheng, Z., Haque, M.M., Wang, M., 2019. A game theory-based approach for modelling mandatory lane-changing behaviour in a connected environment. Transp. Res. C: Emerg. Technol. 106, 220–242.
Arun, A., Haque, M.M., Bhaskar, A., Washington, S., Sayed, T., 2021. A systematic mapping review of surrogate safety assessment using traffic conflict techniques. Accid. Anal. Prev. 153, 106016.
Behnood, A., Al-Bdairi, N.S.S., 2020. Determinant of injury severities in large truck crashes: a weekly instability analysis. Saf. Sci. 131, 104911.
Bezdek, J.C., 2013. Pattern Recognition with Fuzzy Objective Function Algorithms. Springer Science & Business Media.
Chawla, N.V., Bowyer, K.W., Hall, L.O., Kegelmeyer, W.P., 2002. Smote: synthetic minority over-sampling technique. J. Artif. Intel. Res. 16, 321–357.
Chen, Q., Gu, R., Huang, H., Lee, J., Zhai, X., Li, Y.e., 2021a. Using vehicular trajectory data to explore risky factors and unobserved heterogeneity during lane-changing. Accid. Anal. Prev. 151, 105871.
Chen, T., Shi, X., Wong, Y.D., 2019. Key feature selection and risk prediction for lane-changing behaviors based on vehicles' trajectory data. Accid. Anal. Prev. 129, 156–169.
Chen, T., Shi, X., Wong, Y.D., 2020a. A lane-changing risk profile analysis method based on time-series clustering. Physica A: Stat. Mech. Appl.
Chen, T., Shi, X., Wong, Y.D., Yu, X., 2020b. Predicting lane-changing risk level based on vehicles' space-series features: a pre-emptive learning approach. Transp. Res. C: Emerg. Technol. 116, 102646.
Chen, T., Wong, Y.D., Shi, X., Yang, Y., 2021b. A data-driven feature learning approach based on copula-bayesian network and its application in comparative investigation on risky lane-changing and car-following maneuvers. Accid. Anal. Prev. 154, 106061.
Cheng, W., Fei, H., Asad, J.K., 2021. Driver lane-changing behavior prediction based on deep learning. J. Adv. Transp. 2021.
Das, A., Ahmed, M.M., 2021. Exploring the effect of fog on lane-changing characteristics utilizing the shrp2 naturalistic driving study data. J. Transp. Safety Security 13 (5), 477–502.
Fitch, G., Lee, S., Klauer, S., Hankey, J., Sudweeks, J., Dingus, T., 2009. Analysis of lane-change crashes and near-crashes (report no. Dot hs 811 147). National Highway Traffic Safety Administration.
Gu, X., Cai, Q., Lee, J., Xiang, Q., Ma, Y., Xu, X., 2020. Proactive crash risk prediction modeling for merging assistance system at interchange merging areas. Traffic Inj. Prev. 21 (3), 234–240.
Han, T., Jing, J., Özgüner, Ü., Year. Driving intention recognition and lane change prediction on the highway. In: Proceedings of the 2019 IEEE Intelligent Vehicles Symposium (IV), pp. 957-962.
Hou, Y., Edara, P., Sun, C., 2015. Situation assessment and decision making for lane change assistance using ensemble learning methods. Expert Syst. Appl. 42 (8), 3875–3882.
Huang, H., Wang, J., Fei, C., Zheng, X., Yang, Y., Liu, J., Wu, X., Xu, Q., 2020. A probabilistic risk assessment framework considering lane-changing behavior interaction. Science China Information Sciences 63 (9), 1–15.
Hyun, K.K., Jeong, K., Tok, A., Ritchie, S.G., 2019. Assessing crash risk considering vehicle interactions with trucks using point detector data. Accid. Anal. Prev. 130, 75–83.
Joo, Y.-J., Park, H.-C., Kho, S.-Y., Kim, D.-K., 2021. Reliability-based assessment of potential risk for lane-changing maneuvers. Transp. Res. Rec. 03611981211010800.
Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., Liu, T.-Y., 2017. Lightgbm: A highly efficient gradient boosting decision tree. Adv. Neural Inf. Process. Systems 30, 3146–3154.
Kesting, A., Treiber, M., Helbing, D., 2007. General lane-changing model mobil for car-following models. Transp. Res. Rec. 1999 (1), 86–94.
Krajewski, R., Bock, J., Kloeker, L., Eckstein, L., Year. The highd dataset: A drone dataset of naturalistic vehicle trajectories on german highways for validation of highly automated driving systems. In: Proceedings of the 2018 21st International Conference on Intelligent Transportation Systems (ITSC), pp. 2118-2125.
Kumar, P., Perrollaz, M., Lefèvre, S., Laugier, C., Year. Learning-based approach for online lane change intention prediction. In: Proceedings of the IEEE Intelligent Vehicles Symposium, 2013.
Li, K., Wang, X., Xu, Y., Wang, J., 2016. Lane changing intention recognition based on speech recognition models. Transp. Res. C: Emerg. Technol. 69, 497–514.
Li, M., Li, Z., Xu, C., Liu, T., 2020. Short-term prediction of safety and operation impacts of lane changes in oscillations with empirical vehicle trajectories. Accid. Anal. Prev. 135, 105345.
Li, X., Wang, W., Roetting, M., 2018. Estimating driver's lane-change intent considering driving style and contextual traffic. IEEE Trans. Intell. Transp. Syst. 20 (9), 3258–3271.
Liu, S., Wang, X., Hassanin, O., Xu, X., Yang, M., Hurwitz, D., Wu, X., 2021. Calibration and evaluation of responsibility-sensitive safety (rss) in automated vehicle performance during cut-in scenarios. Transp. Res. C: Emerg. Technol. 125, 103037.
Liu, Y., Zhao, P., Qin, D., Yang, Y., Chen, Z., Year. Driving intention identification based on long short-term memory neural network. In: Proceedings of the 2019 IEEE Vehicle Power and Propulsion Conference (VPPC), pp. 1-6.
Mahajan, V., Katrakazas, C., Antoniou, C., 2020a. Crash risk estimation due to lane changing: a data-driven approach using naturalistic data. IEEE Transactions on Intelligent Transportation Systems.

Mahajan, V., Katrakazas, C., Antoniou, C., 2020b. Prediction of lane-changing maneuvers with automatic labeling and deep learning. Transp. Res. Rec. 2674 (7), 336–347.
Martinez, C.M., Heucke, M., Wang, F.-Y., Gao, B., Cao, D., 2018. Driving style recognition for intelligent vehicle control and advanced driver assistance: a survey. IEEE Trans. Intell. Transp. Syst. 19 (3), 666–676.
Meng, Q., Weng, J., 2012. Classification and regression tree approach for predicting drivers' merging behavior in short-term work zone merging areas. J. Transp. Eng. 138 (8), 1062–1070.
National Center for statistics and analysis, 2019. Fatality analysis reporting system (fars) analytical user's manual, 1975-2018 (report no. Dot hs 812 827). National Highway Traffic Safety Administration, Washington, DC.
Oh, C., Park, S., Ritchie, S.G., 2006. A method for identifying rear-end collision risks using inductive loop detectors. Accid. Anal. Prev. 38 (2), 295–301.
Pang, M.Y., Jia, B., Xie, D.F., Li, X.G., 2020. A probability lane-changing model considering memory effect and driver heterogeneity. Transportmetrica B: Transp. Dynamics 8 (1), 72–89.
Park, H., Oh, C., Moon, J., Kim, S., 2018. Development of a lane change risk index using vehicle trajectory data. Accid. Anal. Prev. 110, 1–8.
Sarkar, A., Hickman, J.S., McDonald, A.D., Huang, W., Vogelpohl, T., Markkula, G., 2021. Steering or braking avoidance response in shrp2 rear-end crashes and near-crashes: A decision tree approach. Accid. Anal. Prev. 154, 106055.
Shangguan, Q., Fu, T., Wang, J., Jiang, R., Fang, S.E., 2021a. Quantification of rear-end crash risk and analysis of its influencing factors based on a new surrogate safety measure. J. Adv. Transp. 2021.
Shangguan, Q., Fu, T., Wang, J., Luo, T., Fang, S., 2021b. An integrated methodology for real-time driving risk status prediction using naturalistic driving data. Accid. Anal. Prev. 156, 106122.
Shi, X., Wong, Y.D., Li, M.-F., Palanisamy, C., Chai, C., 2019. A feature learning approach based on xgboost for driving assessment and risk prediction. Accid. Anal. Prev. 129, 170–179.
Traffic Management bureau of the public security ministry, 2020. Annual statistic yearbook of road traffic accidents in china (2019). Beijing.
Wang J., Fu T., 2021 TJRD TS. https://www.tjrdts.com.
Xia, Y., Qu, Z., Sun, Z., Li, Z., 2021. A human-like model to understand surrounding vehicles lane changing intentions for autonomous driving. IEEE Trans. Veh. Technol. 70 (5), 4178–4189.
Xie, D., Fang, Z., Jia, B., He, Z., 2019. A data-driven lane-changing model based on deep learning. Transp. Res. C: Emerg. Technol. 106, 41–60.
Xu, T., Wen, C., Zhao, L., Liu, M., Zhang, X., 2019. The hybrid model for lane-changing detection at freeway off-ramps using naturalistic driving trajectories. IEEE Access 7, 103716–103726.
Yang, M., Wang, X., Quddus, M., 2019. Examining lane change gap acceptance, duration and impact using naturalistic driving data. Transp. Res. C: Emerg. Technol. 104, 317–331.
Yu, B., Bao, S., Feng, F., Sayer, J., 2019. Examination and prediction of drivers' reaction when provided with v2i communication-based intersection maneuver strategies. Transp. Res. C: Emerg. Technol. 106, 17–28.
Yu, B., Bao, S., Zhang, Y., Sullivan, J., Flannagan, M., 2021. Measurement and prediction of driver trust in automated vehicle technologies: an application of hand position transition probability matrix. Transp. Res. C: Emerg. Technol. 124, 102957.
Zhao, L., Rilett, L., Haque, M.S., 2021. Hidden markov model of lane-changing-based car-following behavior on freeways using naturalistic driving data. Transp. Res. Rec. 2675 (8), 550–561.
Zheng, L., Sayed, T., Mannering, F., 2021. Modeling traffic conflicts for use in road safety analysis: a review of analytic methods and future directions. Anal. Methods Accid. Res. 29, 100142.
Zheng, Z., 2014. Recent developments and research needs in modeling lane changing. Transp. Res. Part B: Methodol. 60 (1), 16–32.